

## Summary

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$$\begin{aligned}
& \bullet \frac{d}{dx} [\sin^{-1}(x)] = \frac{1}{\sqrt{1-x^2}} & \bullet \frac{d}{dx} [\cos^{-1}(x)] = \frac{-1}{\sqrt{1-x^2}} \\
& \bullet \frac{d}{dx} [\tan^{-1}(x)] = \frac{1}{1+x^2} & \bullet \frac{d}{dx} [\cot^{-1}(x)] = \frac{-1}{1+x^2} \\
& \bullet \frac{d}{dx} [\sec^{-1}(x)] = \frac{1}{|x|\sqrt{x^2-1}} \text{ for } |x| \geq 1 & \bullet \frac{d}{dx} [\csc^{-1}(x)] = \frac{-1}{|x|\sqrt{x^2-1}} \text{ for } |x| \geq 1
\end{aligned}$$

**Theorem 1** (The Inverse Function Theorem). If  $f$  is a differentiable function that is one-to-one near  $a$  and  $f'(a) \neq 0$ , then:

- (a)  $f^{-1}(x)$  is **defined** for  $x$  near  $b = f(a)$ ,
- (b)  $f^{-1}(x)$  is **differentiable** for  $x$  near  $b = f(a)$ ,
- (c) last, but not least:  $\left[ \frac{d}{dx} f^{-1}(x) \right]_{x=b} = \frac{1}{f'(a)}$  where  $b = f(a)$ .